

ISABELLE STRATTON

[LinkedIn](#)

EDUCATION

Carnegie Mellon University
M.S. Sustainable and Regenerative Design
May 2027

University of Florida
B.S. Mathematics
August 2023

University of Florida
B.F.A. Fine Arts
August 2023

TECHNICAL SKILLS

BUILDING ANALYSIS
Revit (modeling and energy analysis), IESVE (CFD and energy modelling), EnergyPlus, eQUEST, ClimateStudio, Climate Consultant

COMPUTATIONAL DESIGN
Grasshopper, Python (parametric workflows and scripting)

3D MODELING
Rhino, SolidWorks

GEOSPATIAL
ArcGIS ecosystem

VISUALIZATION
Adobe Creative Suite, Microsoft Office

EXPERIENCE

Carnegie Mellon University 2026–Present
Research Assistant Pittsburgh, PA

- Design and prototype sustainable materials from biomaterials and construction waste within a circular-economy framework.
- Support research in material ecologies, exploring bio-based and waste-derived material systems.
- Conduct material characterization and testing to evaluate performance and environmental impact.

The University of Florida 2025–2026
GIS Analyst Gainesville, FL

- Built an ArcGIS tool that maps parcel-level ecosystem service values for conservation planning.
- Processed spatial datasets (parcels, canopy, wetlands, soils) with reprojection, zonal statistics and field calculations.
- Integrated economic valuation models with GIS layers for automated per-parcel outputs; developed a unified geodatabase and an interactive web map for county stakeholders.
- Worked with Dr. Kotryna Klizentyte, who carried out the literature review behind the valuation calculations.

Behavioral Health Services North 2024–Present
Data Analyst Plattsburgh, NY

- Deliver diagnostic and predictive analyses of healthcare needs aligned with CCBHC reporting standards.
- Engineered a risk-prediction model using multivariate statistics and a Random Forest classifier.
- Manage multiple databases in Excel and SQL and perform analytics on protected health information in full HIPAA compliance.

Hulcr Lab 2023
Laboratory Technician Gainesville, FL

- Worked with Christopher Marais on PCA and K-means clustering for an AI-based bark beetle classifier.
- Trained new technicians in sample processing and photography; operated a macro-camera for AI data collection.

PUBLICATIONS

Decision-making tool to value ecosystem services on potential land acquisitions in Hillsborough County, Florida In preparation
Isabelle C. Stratton, Kotryna Kliizentyte Gainesville, FL

- Publication in the works for the Hillsborough County ecosystem service valuation project; to be presented at the Tampa Bay Innovation Summit.

Progress in Developing a Bark Beetle Identification Tool 2025
G. Christopher Marais, Isabelle C. Stratton, Andrew J. Johnson, Jiri Hulcr Gainesville, FL

- Presents a tool for identifying bark beetles, invasive species that damage forests globally and whose uniform, homoplastic morphology makes identification difficult.

PROJECTS

Generational Relief in Prosthetics 2023

- Was one of six in the 3D-Printed Assistive Devices Division; converted digital renderings of assistive prosthetics into 3D models for recipients.
- Troubleshoot and repaired 3D printers and helped adapt toys for movement-restricted children.

Touch-Activated Warming Sandbox 2022

- Wrote Arduino code for touch sensors and 12-volt heating elements, resolved voltage-regulation issues, and built a working prototype for educational and therapeutic use.

AWARDS

- **Robert Long Mathematics Essay Contest, Honorable Mention** for an essay on a-periodic tilings in Islamic mosques (2023)
- **Neilson Scholarship**, with a preparator internship at the Harn Museum of Art; one of the youngest recipients (2020)
- **American Visions Award**, Scholastic Art and Writing, for illustration; one awarded per region nationally (2019)
- **National Gold Medalist**, Scholastic Art and Writing, illustration and mixed media (2018, 2019)
- **Bright Futures Scholar**, full undergraduate tuition (2019–2023)
- **Raimondi Family Photography Scholarship** (2019)
- **Jason Budjinski Memorial Scholarship**, academic excellence (2019)